

Table 1: CPU time on dataset collection 1 with 20.0% constraints (without centroid mutation)

| Constraints Level<br>Method<br>Dataset | 0.2  |        |      |
|----------------------------------------|------|--------|------|
|                                        | KSKM | KSKM_E | PCCC |
| appendicitis                           | 0.00 | 0.00   | 0.00 |
| breast_cancer                          | 0.00 | 0.00   | 0.01 |
| bupa                                   | 0.00 | 0.00   | 0.00 |
| circles                                | 0.00 | 0.00   | 0.00 |
| ecoli                                  | 0.00 | 0.01   | 0.04 |
| glass                                  | 0.00 | 0.01   | 0.05 |
| haberman                               | 0.00 | 0.00   | 0.00 |
| hayes-roth                             | 0.00 | 0.00   | 0.01 |
| heart                                  | 0.00 | 0.00   | 0.00 |
| ionosphere                             | 0.00 | 0.00   | 0.00 |
| led7digit                              | 0.00 | 0.06   | 0.35 |
| monk-2                                 | 0.00 | 0.00   | 0.00 |
| moons                                  | 0.00 | 0.00   | 0.00 |
| movement_libras                        | 0.01 | 0.63   | 3.07 |
| newthyroid                             | 0.00 | 0.00   | 0.01 |
| saheart                                | 0.00 | 0.00   | 0.00 |
| sonar                                  | 0.00 | 0.00   | 0.00 |
| soybean                                | 0.00 | 0.00   | 0.01 |
| spectfheart                            | 0.00 | 0.00   | 0.00 |
| spiral                                 | 0.00 | 0.00   | 0.00 |
| tae                                    | 0.00 | 0.00   | 0.01 |
| vehicle                                | 0.00 | 0.00   | 0.01 |
| zoo                                    | 0.00 | 0.01   | 0.04 |